

AI-Based Phishing Detection Agent

An Intelligent Agent for Detecting Phishing URLs

BSCS — Artificial Intelligence Course Project

1. Introduction

Every day, millions of people in Pakistan use the internet for banking, shopping, and social media. Cybercriminals take advantage of this by creating fake websites that look real, called phishing websites, to steal usernames, passwords, and financial details. This project designs an AI-based agent that automatically checks a URL and tells the user whether it is Legitimate or Phishing, warning them before any damage is done.

2. Problem Statement — Phishing in Pakistan

The Problem

Phishing attacks trick users into clicking fake links (often sent through SMS, WhatsApp, or email) that imitate banks, telecom companies, or e-commerce sites. The fake page looks identical to the real one and asks the user to enter sensitive information.

Why It Is Dangerous

- Pakistani users lose money directly from bank accounts and mobile wallets (JazzCash, Easypaisa, etc.).
- Stolen credentials are used for identity theft and further fraud.
- Many users are not trained to recognize fake links, especially in low digital-literacy areas.
- Businesses lose customer trust when their brand is used in a phishing scam.

How AI Can Help

An AI agent can automatically examine a URL's features (length, symbols, domain age, use of HTTPS, etc.) and instantly classify it as Legitimate or Phishing — much faster and more consistently than manual checking. This gives users real-time protection without needing technical knowledge.

3. Proposed AI Solution

We propose a Phishing Detection Agent that takes a URL as input, extracts key features from it, and uses a trained AI classification model to predict whether the URL is safe or malicious. If the URL is classified as phishing, the agent immediately shows a warning message to the user before they can proceed.

4. Problem Formulation

Initial State: A raw, unchecked URL entered or clicked by the user.

Goal State: The URL is correctly labeled as Legitimate or Phishing, and the user is informed of the result.

Inputs: The URL string, and features extracted from it (domain name, length, special characters, HTTPS usage, IP-based address, age of domain, etc.).

Actions: Extract features → Analyze features → Classify the URL → Generate a warning (if phishing) or allow access (if legitimate).

Possible Outcomes: (1) URL is Legitimate → user proceeds safely. (2) URL is Phishing → user receives a warning and is blocked/advised not to continue.

5. Simple Graphical Diagram

The flow below shows how the agent processes a URL step by step:



6. Environment Creation

The environment in which the AI agent operates consists of the following components:

- User — the person who enters or clicks a URL (e.g., through browser or messaging app).
- URLs — the actual web links being checked, submitted one at a time.
- URL Dataset — a labeled collection of known legitimate and phishing URLs used to train and test the AI model.
- AI Agent — the core program that extracts features and classifies each URL.
- Classification Result — the output (Legitimate or Phishing) shown back to the user.

How the Agent Observes and Decides

The agent 'observes' the environment through the URL string given to it — this is its sensor input. It extracts measurable features from the URL (such as length, presence of '@' or '-' symbols, IP address instead of domain name, HTTPS presence, and domain age). These features are passed to a trained classification model, which compares patterns learned from the URL dataset and decides the most likely class. The decision (its action) is then delivered to the user as a result and, if needed, a warning.

7. PEAS Analysis

PEAS Element	Description
Performance Measure	Accuracy of classification, low false alarms, fast response time, successful warning of real phishing URLs.
Environment	The internet / browser context where URLs are submitted, along with the URL dataset used for reference.
Actuators	Displaying the result (Legitimate/Phishing) and showing a warning message or blocking the page.
Sensors	The input URL text and its extracted features (domain, symbols, HTTPS status, etc.).

8. Environment Properties

Property	Type	Reason
Observability	Partially Observable	The agent only sees the URL and its extractable features, not the full intent or content behind the website.
Determinism	Deterministic	For the same URL and same features, the agent gives the same classification every time.
Episodic / Sequential	Episodic	Each URL is checked independently; the decision on one

Property	Type	Reason
		URL does not depend on previous URLs.
Static / Dynamic	Dynamic	New phishing websites appear constantly, so the environment keeps changing over time.
Discrete / Continuous	Discrete	The output has a fixed number of classes: Legitimate or Phishing.
Agents	Single-agent	Only one AI agent performs the URL classification task in this system.

9. Simple Example

Example URL: <http://secure-hbl-verify.account-update.info/login>

Step-by-step:

- The agent receives this URL as input.
- It extracts features: unusual long domain name, contains the words 'secure' and 'verify' (common phishing tricks), no official HBL domain, no valid HTTPS certificate.
- The classification model finds these features match known phishing patterns.
- Decision: Phishing.
- The agent immediately shows a warning: " ⚠ Warning: This link looks like a phishing website. Do not enter your bank details."

For comparison, a genuine link such as <https://www.hbl.com/login> would be analyzed and classified as Legitimate, allowing the user to continue safely.

10. Conclusion

Phishing is a growing threat to internet users in Pakistan, especially with the rise of online banking and mobile payments. The proposed AI-Based Phishing Detection Agent offers a simple but effective solution: it automatically analyzes URLs, classifies them as Legitimate or Phishing, and warns users in real time. This kind of intelligent agent can significantly reduce the risk of users falling victim to online scams.